KATLEGO BARAYI

Software Engineer

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GitHub | LinkedIn | Portfolio

PROFESSIONAL SUMMARY

Software Engineer with 1.5 years of professional experience building enterprise CMS platforms, RESTful APIs, and observability tooling in TypeScript at MTN Group's Centre of Excellence. Specialised in Directus extension development, Datadog instrumentation (agentless APM tracing, structured logging), Okta and OAuth 2.0 authentication, and test driven development with a 500+ test Vitest suite. Also experienced with Vue 3, React, Node.js, and Python, plus Docker based deployments and CI/CD pipelines (GitHub Actions, GitLab CI). Certified SAFe Scrum Master with experience in Agile/Scrum environments alongside cross functional teams and stakeholders.

PROFESSIONAL EXPERIENCE

Software Engineer January 2025 to March 2026

MTN Centre of Excellence, Johannesburg

Contributed across three teams within MTN's Centre of Excellence: CMS Core (platform engineering), Hive (HR content management), and MTN One (CMS production support), serving content operations across MTN's pan African footprint.

CMS Core Team

- Architected and built a Datadog observability extension for Directus, implementing agentless APM tracing, structured logging via Winston, secrets management backed by a database table, HTTP request middleware, and change tracking audit hooks for data mutations.
- Implemented enterprise authentication and authorization flows using OAuth 2.0 and Okta, ensuring compliance with organisational security standards across all CMS access points.
- Developed a comprehensive Vitest test suite with 500+ unit and integration tests, significantly improving code coverage and reducing regressions across the CMS extension ecosystem.
- Designed and developed a scalable CMS platform using Directus and TypeScript, enabling flexible content modeling and delivery across consumer facing applications for 10+ internal stakeholders.
- Built 5+ custom Directus extensions integrating internal and external REST APIs, streamlining complex business workflows and reducing manual content operations by an estimated 30%.
- Refactored a monolithic Vue 3 video file interface component into a modular architecture (types, composables, utils), improving maintainability and enabling independent testing of each module.
- Maintained and improved CI/CD pipelines (GitHub Actions, GitLab CI, Docker) supporting reliable automated deployments, reducing release cycle friction and minimising downtime.
- Instrumented PostHog analytics across multiple frontend modules, enabling data driven decisions on content editor workflows and platform usability improvements.

Hive Team (HR)

- Built and delivered an HR focused CMS powering the Hive platform, enabling management of employee records, quick links, company news, upcoming events, theme of the week, Viva Engage content, articles, training resources, Human Capital Management (HCM), and IT service information.

MTN One Team (CMS Support)

- Provided Tier 1 CMS production support, troubleshooting incidents and enabling content editors, improving system usability and stakeholder satisfaction.
- Applied Agile/Scrum principles as a Certified SAFe Scrum Master, facilitating sprint planning, daily stand ups, retrospectives, and backlog refinement.

Volunteer Coding Mentor September 2024 to June 2025

MTN Group, Johannesburg

- Volunteered to lead coding workshops with high school students, teaching programming fundamentals and sharing career insights from WeThinkCode_ to professional software engineering at MTN.
- Mentored learners on problem solving approaches and introduced them to pathways in technology, encouraging participation in STEM fields.

Prototype Developer June 2024 to October 2024

Lelapa AI, Johannesburg

Joined post hackathon to build a production ready Sign Language Interpreter prototype, translating South African Sign Language (SASL) into all 11 of South Africa's official languages.

- Designed and developed an AI powered Sign Language Interpreter in Python, integrating Lelapa AI's NLP language APIs to deliver real time text and audio output, improving accessibility for deaf and hard of hearing users.
- Integrated multilingual translation pipelines supporting dynamic user selected output across all 11 official South African languages, expanding the solution's real world reach.
- Built intuitive user facing React interfaces enabling seamless language and format selection, resulting in a polished prototype recognised as a hackathon finalist.
- Collaborated cross functionally with AI engineers, UX designers, and product stakeholders to rapidly iterate features and validate technical concepts under tight deadlines.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, Java, Dart, HTML5, CSS3, SQL

Frontend: React, Vue 3, Flutter, responsive design, component driven UI development

Backend & Enterprise Development: RESTful API development, Node.js, Express, Spring Boot, FastAPI, API integration, authentication & authorization, enterprise platform development

Databases: SQL, relational database design, query optimization, database integrations, PostgreSQL, SQL Server fundamentals

CMS & Auth: Directus, OAuth 2.0, Okta, enterprise authentication and authorization

Cloud & DevOps: Docker, GitHub Actions, GitLab CI/CD, automated deployment pipelines, cloud deployment workflows, Azure exposure

Observability: Datadog (APM tracing, telemetry, structured logging, instrumentation), PostHog analytics

Testing: Unit testing, integration testing, Vitest, test automation, UAT, code coverage improvement

Tools: Git, GitHub, GitLab, Postman, VS Code, Winston

Methodologies: Agile, Scrum (Certified SAFe Scrum Master), SAFe, sprint planning, backlog refinement

EDUCATION

National Certificate in IT Systems Development (NQF Level 5)

WeThinkCode_, Johannesburg

CERTIFICATIONS

- Certified SAFe Scrum Master
- WeThinkCode_ GenAI Course for Software Engineers

AWARDS & RECOGNITION

- Winner | Nerdma Systems Hack AI thon 2024: AI for Loss Prevention (1st place)
- Finalist | Lelapa AI Hackathon 2024: South African Sign Language Interpreter